

Technology Stack

Date	15 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Technology Stack Details

S. No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	HTML, CSS, JavaScript, React.js	Provides an interactive and user-friendly interface for farmers to input soil data and view crop recommendations. React ensures fast rendering and reusable components.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python (Flask)	Handles data processing, API requests, and integrates the machine learning model for prediction. Flask is lightweight and suitable for ML-based applications.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	MySQL	Stores user data, soil records, and prediction history efficiently with structured query support.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	AWS Render	Used to deploy and host the application online with scalability and easy access from anywhere.

5	Version Control & CI/CD <i>(e.g., GitHub, GitLab, Jenkins)</i>	Git & GitHub	Used for code management, version tracking, and collaboration. Enables continuous integration and deployment.
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	Weather API, Pandas, NumPy, Scikit-learn	Weather API provides environmental data. Pandas and NumPy are used for data processing. Scikit-learn is used to build and train the crop prediction model.